MIL-J-35050C
10 September 1985
SUPERSEDING
MIL-J-35050B
2 February 1967

MILITARY SPECIFICATION

JUICE, GRAPEFRUIT, INSTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers instant grapefruit juice for use by the Department of Defense as an item of general issue and as a component of operational rations.

1.2 Classification. The product shall be of the following styles, grades and classes, as specified.

Style I - Unsweetened - Deleted (see 6.3)
Style II - Sweetened

U.S. Grade A
U.S. Grade B - Deleted (see 6.3)

Class 1 -Unfortified
Class 2 -Fortified - Deleted (see 6.3)

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8915

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SPECIFICATIONS

FEDERAL

TT-C-495	- Coatings, Exterior, for Tinned Food Cans
PPP-C-29	- Canned Subsistence Items, Packaging and Packing of
PPP-C-96	- Cans, Metal, 28 Gage and Lighter
PPP-B-636	- Boxes, Shipping, Fiberboard

MILITARY

MIL-L-1497	- Labeling of Metal Cans for Subsistence Items
MIL-L-35078	- Loads, Unit: Preparation of Nonperishable Subsistence Items; General Specification for
MIL-D-43266	- Desiccant and Desiccation, Method of, for Packaging Subsistence

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

Laws and Regulations

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder (21 CFR parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

U.S. Standards for Grades of Grapefruit Juice

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(Application for copies should be addressed to the Chief, Processed Products Branch, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

U.S. Standards for Condition of Food Containers

(Application for copies should be addressed to the Director, Market Research and Development Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Ave., N.W., Washington, DC 20418).

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

AMERICAN DRY MILK INSTITUTE, INC.

Standards for Grades of Dry Milk, Including Methods of Analysis, Bulletin 916

(Application for copies should be addressed to the American Dry Milk Institute, Inc., 130 North Franklin Street, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

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Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Suite 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent and insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.1.1 Grapefruit juice concentrate. Grapefruit juice concentrate shall comply with the requirements for U.S. Grade A of the U.S. Standard for Grades of Concentrated Grapefruit Juice for Manufacturing, except that the Brix-acid ratio shall be not less than 7.5 to 1 and not greater than 14 to 1. The concentrates shall have been prepared in high vacuum, low temperature, falling film concentrators or in equipment such as Thermally Accelerated Short Time Evaporators which produce equivalent material to insure avoidance of heat damage, and shall be placed in freezer storage immediately after processing and frozen to temperatures below 0°F within 4 days, or may be dehydrated immediately without prior freezing, provided holding time, temperature and conditions are such that no deteriorative changes occur in the concentrate.

3.1.2 Oil of grapefruit. Oil of grapefruit shall be cold-pressed grapefruit oil which meets the requirements for expressed (cold-pressed) grapefruit oil specified in the Food Chemicals Codex, and is free of a turpentine-like odor, a partially deterpened oil prepared therefrom or a combination of both. Oil of grapefruit shall be added directly to the grapefruit juice concentrate immediately prior to dehydration or may be added in the form of readily soluble entrapped grapefruit oil granules. The entrapped grapefruit oil granules shall have been prepared by entrapping oil of grapefruit in sucrose, dextrose, corn syrup, corn syrup solids, gum acacia, other suitable vegetable gums, glycerol, and emulsifiers or any combination thereof.

3.1.3 Sweetening ingredient. Sweetening ingredient shall be white, refined granulated cane or beet sugar, or combinations thereof, or in combination with white refined dextrose or corn syrup solids which provide equivalent sweetness.

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3.1.4 Sodium sulfite or sodium bisulfite. Sodium sulfite or sodium bisulfite shall meet the applicable requirements of the Food Chemicals Codex and shall be added to the grapefruit juice concentrate prior to dehydration, in such quantity that the sulfur dioxide content of the dehydrated product shall meet the requirements specified in 3.3.

3.1.5 Drying aids. Drying aids such as foam stabilizers, vegetable gums or carboxymethylcellulose, which have been approved by the Food and Drug Administration, may be used in quantities permitted by that agency.

3.2 Processing. Preparation of dehydrated grapefruit juice shall be in accordance with a method or methods which insure maximum preservation of the natural characteristics of grapefruit juice. Sweetening ingredients shall be added in the proportion of 2 ounces of sweetening ingredient solids to 13.5 ounces of grapefruit juice solids either prior to or after dehydration.

3.2.1 Processing equipment. All machinery, equipment, and parts thereof, with which liquid materials may come into contact during processing, shall comply with 21 CFR Part 110 current good manufacturing practice.

3.3 Finished product. The finished product shall comply with the applicable requirements for U.S. Grade A, including a flavor not lower than U.S. Grade A, as specified for dehydrated grapefruit juice in the U.S. Standards for Grades of Grapefruit Juice, except that the color will be allowed a U.S. B grading, and the sulfur dioxide content shall be not less than 100 parts per million (ppm). The moisture content shall not exceed 2.5 percent at time of pack, and shall not exceed 1.5 percent within 21 days after packaging. Lumps of material which are fragile in nature and can be readily broken up by shaking or by application of light pressure shall not be considered as defects. The finished product shall contain not less than 13.5 avoirdupois ounces of grapefruit solids plus 2 avoirdupois ounces of sweetening ingredient(s) solids per gallon of juice.

3.4 Plant qualification. The product shall be prepared, processed and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Processing, Packaging or Holding of Human Foods", and the plant sanitation requirements of the appropriate Government inspection agency.

3.5 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4 2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of supervision necessary to assure compliance with the requirements of this document.

4 3 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4 3 1 Component and material inspection. In accordance with 4 1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified or qualified in this document or applicable purchase document.

4 3.1 1 Ingredient and component examination Conformance of ingredients and components to identity, condition and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be evident by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses or other such supplier-furnished valid documents acceptable to the inspection agency. In addition, prior to use, each ingredient shall be examined organoleptically, as necessary, to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient lot or component lot or of any involved product.

4.3.1.2 Unfilled can inspection. Conformance of unfilled cans to the requirements specified in 5.1.1 shall be determined by examination of certificates of conformance or of other valid documents. Any nonconformance shall be cause for rejection of the can lot or of any involved product.

4.3.2 In-process examination. In-process examination shall be performed to determine conformance to the processing, desiccant addition, and unit packing requirements. Any nonconformance revealed by actual examination or by review of records of calculation of desiccant amounts or by review of other valid documents shall be cause for rejection of the involved product.

4.3.3 Moisture determination at time of pack. Eight cans of product shall be randomly selected from the lot at time of pack. The moisture content of the product from each can shall be determined in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists; Chapter: Sugar and Sugar Products, Method; Moisture in Sugars, Vacuum Drying; except that the temperature shall be 60°C and a permitted alternative to redrying every hour is to dry for a consecutive 16 hours, cool in a desiccator, and weigh. Any failure to conform to the moisture requirement in 3.3 shall be classified as a major defect and shall be cause for rejection of the lot.

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4.3.4 Moisture determination within 21 days after packaging. Eight cans of product shall be randomly selected from the lot after packaging. The moisture content of the product from each can shall be determined within 21 days after packaging in accordance with 4.3.3. The contactor shall indicate the minimum length of time within the 21 day period that the cans are to be held prior to testing. Any failure to conform to the moisture requirement in 3.3 shall be classified as a major defect and shall be cause for rejection of the lot.

4.3.5 Net weight examination on a dry solids basis. Examination shall be made during packaging to determine compliance with the fill requirement of 5.1.1. Any individual sample unit having a net weight on a dry solids basis of less than 15.5 ounces shall be scored as a minor defect. The moisture content of the composite (taken at time of packaging and tested in accordance with 4.3.3) representing the lot in question shall be used to make this determination. If moisture test results are not available at time of packaging, reserve computations until test results become available. Net weight results shall be recorded to the nearest 0.1 ounce. The lot size shall be expressed in units of cans. The sample unit shall be one filled can without desiccant. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

4.3.6 Product examination. The finished product shall be examined for the defects listed in table I. The lot size shall be expressed in cans. The sample unit shall be the contents of one filled and sealed can. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for minor defects.

TABLE I. Product defects 1/

Category		Defect
Major	Minor	
		Desiccant
101		Desiccant bag missing.
102		Desiccant bag has tear, hole, or open seal.
	201	Desiccant bag folded over itself.
		Product
103		Lumps of product which cannot be readily broken up by shaking or by application of slight finger pressure.

1/ The presence of any foreign material (e.g. glass, dirt, insect parts,

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hair, wood, metal, etc.), foreign odor or flavor (e.g. burnt, scorched, moldy, rancid, sour, stale, etc.), or foreign color shall be cause for rejection of the lot.

4.3.7 Oxygen in headspace testing (applicable to gas packed cans). The filled and sealed cans shall be tested for oxygen in headspace in accordance with the Determination of Oxygen method in Bulletin 916 of the American Dry Milk Institute, Inc. Alternatively, the headspace oxygen content may be determined by an Instrumentation Laboratories Oxygen Analyzer or equivalent instrument. Test results shall be reported to the nearest 0.1 percent. Any result failing to conform to the oxygen in headspace requirements in 5.1.1 shall be classified as a major defect. When referee testing is necessary, The American Dry Milk Institute Method shall be followed. The lot size shall be expressed in cans. The sample unit shall be one filled and sealed can. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.3.8 Sulfur dioxide content testing. The finished product shall be tested for sulfur dioxide content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists; Chapter: Food Additives; Direct, Method: Sulfurous Acid (Total) in Food; Modified Monier - Williams Method. Test results shall be reported to the nearest 1 ppm. The lot size shall be expressed in cans. The sample shall be a 1-pound composite derived from the number of cans indicated by inspection level S-2. Any result failing to conform to the sulfur dioxide content requirements in 3.3 shall be classified as a major defect and shall be cause for rejection of the lot.

4.3.9 Can condition examination. Examination of filled and sealed cans shall be in accordance with the United States Standards for Condition of Food Containers, except that inspection for labeling shall be in accordance with MIL-L-1497 (see 5.4).

4.3.10 Leakage and vacuum inspection. Inspection for leakage and vacuum shall be in accordance with PPP-C-29 (see 5.1.1).

4.3.11 Shipping container examination. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in assembly, closure and reinforcement shall be in accordance with the appendix of PPP-B-636. In addition, the following defects shall be classified as follows:

Major: National stock number, item description, contract number or date of pack markings missing, incorrect, or illegible.
Number of cans not as specified.
Reinforcement with nonmetallic strapping or tape is not used.

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Minor: Other required markings missing, incorrect, or illegible.
Tiered cans not separated as specified for Level A packing.
Container not snug-fitting.
Arrangement of cans not as specified.

Level C shipping containers shall be examined for the marking, arrangement, and number of cans defects specified above.

4.3.12 Unit load inspection. Inspection of unit loads shall be in accordance with quality assurance provisions of MIL-L-35078.

5. PACKAGING

5.1 Preservation. The product shall be preserved in accordance with level A or C, as specified (see 6.1).

5.1.1 Level A. Not less than 15.5 avoirdupois ounces of the finished product, on a dry solids basis, shall be filled into a size 401 by 411 or 401 by 602 open top style, round, metal can with soldered or welded side seam and compound-lined, double-seamed ends. The can shall be made throughout from not less than commercial 0.25-pound electrolytic tin plate per base box and shall be coated on the outside with a coating conforming to type I of TT-C-495. Each can shall be provided with sufficient desiccant complying with MIL-D-43266 to reduce the moisture content to 1.5 percent. The amount of desiccant required shall be calculated from the formula specified in MIL-D-43266 except that the value for percent water as used in the formula shall be the actual percent moisture content of the product, minus 1.5 percent. A record of the calculations shall be maintained. After filling, the can shall be hermetically sealed under conditions which will result in a vacuum of not less than 15 inches of mercury, when tested in accordance with the method specified in PPP-C-29. Alternatively, the product shall be unit packed under an atmosphere of nitrogen so that the oxygen content of the gases in the filled and sealed container do not exceed 2.0 percent when tested in accordance with 4.3.6. Gas-packed cans shall not show leakage when tested in accordance with PPP-C-29.

5.1.2 Level C. The product shall be unit-packed as specified in 5.1.1, except that cans with or without commercial exterior coating will be acceptable. Alternatively, cans may be made from 0.20 pound per base box electrolytic tin plate with an exterior commercial coating.

5.2 Packing. Twenty-four cans of the product, arranged 4 in length, 3 in width and 2 in depth, shall be packed on end in a snug-fitting shipping container complying with level, A, B, or C, as specified (see 6.1).

5.2.1 Level A packing. Each shipping container shall be a snug-fitting fiberboard box, constructed and closed in accordance with style RSC, grade V2s of PPP-B-636. Tiers of cans shall be separated by a fiberboard pad. Each

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fiberboard box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament-reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. The shipping container shall be a fiberboard box, constructed and closed in accordance with style RSC, V3c, V3s, or V4s, of PPP-B-636. Each fiberboard box shall be reinforced with nonmetallic strapping or pressure-sensitive filament-reinforced tape in accordance with the appendix of PPP-B-636.

5.2.3 Level C packing. The shipping container shall be in accordance with the National Motor Freight Classification or Uniform Freight Classification, as applicable.

5.3 Unit loads. When specified (see 6.1), the product, packed as specified in 5.2.2 and 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping.

5.4 Labeling and marking.

5.4.1 Cans. Cans shall be labeled in accordance with MIL-L-1497. In addition, the following directions for use shall be included on the label:

JUICE, GRAPEFRUIT, INSTANT

GAS PACKED (when applicable)

Preserved with sulfur dioxide

Directions for use

IMPORTANT: DISCARD DESICCANT BAG

Discard Desiccant Bag. Pour approximately a quart of cold water into a gallon measure. While whipping water, gradually add product. When product has dissolved, add water to make one gallon of juice and stir. If desired, 1/4 cup of sugar may be added. Makes sixteen 8 fl. oz. servings of juice. Alternatively, when preparing juice in quantity, add contents of each can to 3-3/4 quarts of cold water and stir briskly.

5.4.2 Shipping containers. The shipping containers shall be marked in accordance with MIL-STD-129.

5.4.3 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

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6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this document.

(b) Levels of preservation and packing required (see 5.1 and 5.2).

(c) Type and class of unit load when unit loading is required (see 5.2.1 and 5.3).

6.2 Appropriate level of pack. Based on conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 Style, Grade and Class. Style I (Unsweetened), U.S. Grade B and Class 2 (Fortified) have been deleted since they are no longer required.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing Activity:

Army - GL

Project No. 8915-0366

Review activities:

Army - MD, TS
Navy - MC, MS
DP - SS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-J-35050C	JUICE, GRAPEFRUIT, INSTANT

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 20 Nov 84

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NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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